

KARTHIKEYA BHAMIDIPATI

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Career Objectives

Detail-oriented **Computer Science** graduate pursuing a **Master's in Data Science and Spatial Analytics** with expertise in **Artificial Intelligence, GeoAI, Remote Sensing, and Geospatial Data Analytics**. Skilled in developing end-to-end AI applications involving **deep learning, computer vision, spatial data processing, and full-stack software development** using **PyTorch, TorchGeo, FastAPI, Next.js**, and modern GIS technologies. Passionate about building scalable geospatial intelligence platforms for **Earth Observation, environmental monitoring, and urban analytics** through research-driven innovation and robust software engineering.

Education

M.Sc. Data Science and Spatial Analytics (8.03 CGPA)

2025 – Present

Symbiosis Institute of Geoinformatics, Pune

Relevant Coursework: Computer Vision, Machine Learning, Geospatial analytics, Google Earth Engine, ArcGIS Pro, Cybersecurity

B.Sc. (Hons.) Computer Science (8.12 CGPA)

2025

Sri Sathya Sai Institute of Higher Learning, Muddenahalli

Relevant Coursework: Data Structures, Operating Systems, DBMS, Computer Networks, Software Engineering

Intermediate (94.3%)

2022

Sri Chaitanya Junior kalasala [TSBIE]

Secondary Education (10.0 GPA)

2020

P N M high School [TSBSE]

Skills and Certifications

Programming Languages

C, C++, Python, Java, HTML, CSS, JavaScript, LaTeX, PL/pgSQL

Geospatial Tools

Google Earth Engine, Google Earth Pro, ArcGIS Pro, QGIS, TorchGeo, Rasterio

Frameworks & Libraries

ReactJS, Django, ExpressJS, NodeJS, Spring Boot, Material-UI (MUI)

Databases

MySQL, MongoDB, PostgreSQL

Developer Tools

Git, GitHub, Visual Studio Code, Jupyter Notebook

Design & Multimedia

Canva, DaVinci Resolve

Deployment & Hosting

Vercel, Render, Microsoft Azure

Certifications

Introduction to CSS3 – Umich (Feb 2024)

Introduction to HTML5 – Umich (Dec 2023)

Introduction to Cyber Attacks – NYU (Jun 2023)

Introduction to Cybersecurity – Cisco (Oct 2024)

Introduction to Data Science – Cisco (Jun 2023)

Data Analytics Essentials – Cisco (Sep 2024)

Web Development Bootcamp – Udemy (Sep 2024)

Introduction to model context protocol – Anthropic (Oct 2025)

Big Data Foundations – IBM (April 2026)

Experience

Software Engineer Intern

CYIENT May 2025 – July 2025

Designed and implemented a Tkinter-based GUI integrated with an OCR pipeline (Azure AI with tailored prompts) to extract chute-liner specifications with over 98% accuracy. Automated end-to-end Excel reporting for the sentiment analysis project using Pandas and XlsxWriter, reducing manual effort by 80% and enabling one-click dashboard generation. Conducted extensive manual testing of the chute-liner extraction workflow, including prompt engineering to optimize output accuracy. Performed thorough validation for the ZIPLY Excel workflows, ensuring accurate extraction and verification of T-blocks and associated information.

Leadership & Activities

Technical Head - Nirvai Club

Led AI-focused initiatives, hackathons, and collaborative research-driven projects.

Placement Team Member - Symbiosis

Managed LinkedIn/Meetup outreach and coordinated with industry professionals for placements.

Vice-President - Digital Enlightenment Club

Conducted computer literacy programs for rural students, improving digital accessibility.

Student Internship Coordinator - IIC

Streamlined internship processes and facilitated student-industry engagement.

Leader - Eco-Einsteins Club

Led sustainability initiatives, reducing paper usage by 20% and improving waste management.

Editor - Multimedia Department

Produced 7+ videos, increasing campus engagement by 60%.

Team Lead - Insight Department

Led a 5-member team to publish campus magazine, increasing student contributions by 30%.

CHATTY MERN Stack with Socket.io

[GitHub]

Engineered a real-time messaging web app with socket-based communication. Increased user engagement by 30%. Deployed frontend on Vercel and backend on Render.

HSBC Mini-Project C++

[GitHub]

Developed a secure hostel management system using file handling and role-based access control. Focused on modularity, data integrity, and restricted access.

CONTRLZ React, Spring Boot, MQTT, ESP01, 5V Relay

[GitHub]

Built an IoT-based hostel automation system enabling real-time control of lights and fans through a web dashboard. Enhanced energy efficiency and user convenience.

Workshops, Training and Hackathons

HERE Technologies Geospatial Hackathon

Developed a geospatial intelligence system for POI change detection in Singapore using OpenStreetMap as baseline and integrating multi-source data (TomTom, Foursquare). Implemented data standardization, spatial indexing, and semantic matching with AI-based scoring to detect new, closed, and anomalous locations. Gained hands-on experience in real-world geospatial pipelines, API integration, and rapid prototyping under time constraints.

GDG DevFest pune 2025

Attended GDG DevFest 2025, gaining practical insights into AI agents, offline LLMs, Google ADK, and agent-driven architectures. Learned fundamentals of Flutter for cross-platform UI development and system design for AI solutions. Built awareness of responsible automation, human-centered AI, and emerging GenAI tools and workflows.

Projects

GeoSentinel AI – Deep Learning for Remote Sensing

[GitHub]

Building an end-to-end geospatial intelligence platform integrating Sentinel-2 Earth Observation data, U-Net/DeepLabV3+, TorchGeo, FastAPI, and GIS analytics for land-cover segmentation, temporal change detection, benchmark evaluation, and interactive visualization.